

أبحاث مُريد • مدونة استئناف التراث الفكري والروحي لإنسان ٢٠٢٦

Machine research M1–M6 • Moried

وثيقة بحثية من أوراق المشروع • تُنشر كما هي للاطلاع

Machine research — M1–M6

Lab run 2026-07-17 per RESEARCH_BRIEF_mervin_2026-07-17.md · evidence-linked · items marked (unverified) or (analysis-based) where applicable

M1 — Knowledge representation

Approach	Faithfulness	Weekend-feasible	Ceiling
Framework-in-system-prompt (structured markdown ontology)	High for methodology; degrades under multi-turn pressure (SysBench)	Highest	Context rot — keep ≤ ~10k tokens; breaks past ~50–100 ailment entries
RAG over source texts	High for doctrine — grounding cuts hallucination (MIND-SAFE , JMIR)	High	Unbounded corpus; ceiling = retrieval quality
Knowledge graph	Best multi-hop fidelity (OntologyRAG)	Low	Overkill below hundreds of entities
Fine-tuning	Poor for facts — RAG beats FT (Ovadia , EMNLP 2024 ; FineTuneBench)	None	Voice/register only, never doctrine

Production mental-health bots converge on **structure-first hybrids**: Woebot clinician-scripted ([why](#)); Wysa runs LLMs inside a rule-based frame ([World Psychiatry](#)); Youper's thin GPT layer was pulled from stores 04/2026 ([Common Sense Media](#)).

v0 recommendation — two-layer split: methodology in prompt, doctrine by retrieval. System prompt ~7–9k tokens, markdown with fixed field schema ([format matters](#), [He et al.](#)): ~20 ailment cards ≈150 tok each (name/ar · markers · discriminators-vs-siblings · root · cures[] · source-ref) + 7-obstacle stage map (~500) + states vocabulary (~600) + short numbered behavior contract (~1.5k) + 2 worked intake dialogues (~1.6k). **Hard rule: quote scripture/classical text ONLY from retrieved/curated text, never from parametric memory** — the single biggest lever against hallucinated doctrine. Growth: corpus into Vectorize RAG; ailment cards to D1 when >50; fine-tune only ever for register.

M2 — Diagnostic engine

Prior art:

- State-machine phases beat free chat: [AMIE, Nature 2025](#) — history-taking → differential → follow-up, per-turn scratchpad (summarize → update differential → draft).
- Naive "ask questions" prompting HURTS; confidence-gated asking works: [MediQ, NeurIPS 2024](#) — abstention module deciding ask-vs-answer improved accuracy 22.3%.
- Discriminating-question selection: [Ada Health](#) — each answer picks the next question (information gain), keeping totals small.
- Intake = classify concern then route: Wysa ([flow](#)).
- Reflect-then-ask raises readiness + empathy: [MIBot, JMIR](#).
- Socratic scaffolds risk pathologizing normal emotion: [CRBot](#).

Recommended loop: complaint → hidden JSON scratchpad (top-3 candidate ailments + confidence + unknown discriminators; never surfaced) → max 3–4 questions, one per turn, chosen to split top-2 candidates (question bank carries distinguishes: metadata), open behavior-observable phrasing, MI reflection prepended → stop when one candidate dominates (confidence threshold), or budget exhausted (say so honestly, offer both readings), or crisis cues (skip to handoff) → prescribe AS HYPOTHESIS («الملاح دي شبهه») + one cure step + one retrieved quote → write state.

Failure modes to design against:

- Leading questions / sycophancy — models agree with user framing ~58% ([SycEval](#)); rule: question never contains the hypothesis word; ≤1 closed confirmation per intake.
- Over-diagnosis — every differential MUST include a null candidate («لا علة — تعب طبيعي»); abstention is a permitted terminal state (MediQ).
- **Wiswas amplification — the domain killer:** always-available reassurance functions as a compulsion ([Reassurance Robot](#); [NOCD](#)). Rules: same-question loop counter, refuse worship-validity certainty rulings, cap reassurance at ONE then name the pattern and redirect to uncertainty-tolerance + human scholar.
- Multi-turn drift — re-assert phase state in each turn's scratchpad ([SysBench](#)).

M3 — Journey memory (privacy-first)

Threat model: tazkiyah log = religious + health special-category data ([GDPR Art 9](#)). Sector record: [Muslim Pro→X-Mode→military](#) · [Mozilla: prayer/mental apps "fail spectacularly"](#) · [BetterHelp \\$7.8M FTC](#) · [Crisis Text Line/Loris](#). Positive template: [Wysa anonymous-by-design](#).

Store (structured state ~1KB/user)	Never store
Anonymous random ID (no phone/email/name)	Verbatim sin descriptions / confessions
stage (1–7) · ailment_code (fixed taxonomy)	Raw transcripts beyond live session
Active plan: exercise IDs + tier + schedule	Free-text diagnosis narratives
relapse_count · longest_gap · last_checkin	Third-party names
crisis_flag + handoff marker	Location, contacts, device IDs, ad SDKs

Retention pattern: summarize-then-discard into codes. Transcript lives only in request context; session close emits schema-validated JSON diff; no message bodies in logs. In-chat «امسح كل حاجة» → hard delete ([Art 17](#)) + 90-day inactivity TTL.

M4 — Plans engine

- Deterministic templates, LLM fills slots only — generative freedom broke NEDA's Tessa ([CNN](#)).
- Exercises as tiny-habit recipes «بعد [مرساة], هعمل [فعل صغير]» per [Fogg B=MAP](#) — **the five prayer times are free reliable anchors, a structural advantage of this domain.**
- If-then implementation intentions: d=0.65 across 94 tests ([Gollwitzer & Sheeran](#)).
- Weekly check-in = [JITAI](#) decision point; relapse_count + self-report = tailoring variables.
- **Relapse response (fixed template):** normalize (missing once doesn't break formation — [Lally 2010](#)) → re-attribute to trigger not self (avert [abstinence violation effect](#)) → one learning question ([I Am Sober pattern](#)) → step plan DOWN one tier; never penance-exercises.
- **No streaks over sins.** Count recoveries + time-between-falls as the progress metric. (Duolingo streak-freeze churn figure exists but is vendor-reported, unverified.)

M5 — Architecture + guardrails

Client

- L0 edge: anon token · rate limit · input scrub
- L1 safety pre-pass (BEFORE LLM): crisis + wiswas detection
Arabic regex list + tiny classifier · hit ⇒ deterministic template
(validate + 08008880700 + human handoff) · LLM bypassed
- L2 phase FSM IN CODE: intake → diagnose → plan → companion
code owns transitions; LLM cannot switch phase
- L3 LLM: per-phase system prompt + structured state (no transcript history)
- L4 output guard (AFTER LLM): independent classifier on the reply
fatwa-format / takfir / diagnosis / absolution ⇒ regenerate or template
- L5 state writer: schema-validated JSON diff only → D1

- System prompt is NOT a security boundary ([OWASP LLM01](#)); every hard ceiling lives in L1/L4 code+classifier, merely restated in prompt. Tessa's rules lived in the generative layer and broke ([CBS](#)).
- L4 = [Llama Guard pattern](#): independent safeguard model, own taxonomy, updatable separately.
- Crisis pathway: SOS always visible + free-text detection (Wysa caught [82% via AI](#); [peer-published safety design](#)); crisis_flag suppresses plan-talk 72h.
- FSM-over-agents is the industry recommendation for predictability ([Anthropic: building effective agents](#); [Rasa CALM](#)).

M6 — The tongue (voice guide seed)

Target sound: a well-read older brother at the kitchen table — السكران's inner-state precision, الفيغي's tenderness, مجاهد's next-to-you intimacy, شرقاوي's calm, ابن عطاء الله's compression.

Ten rules (each with do/don't in the full agent prompt):

1. (السكران on الحمدان) «وكانه يتكلم عني» — mirror the inner state so he says «شخص قبل ما تنصح».
2. الآية تجربة، مش دليل إدانة — verse woven as lived experience, never proof-text at the user.
3. واحد بيكلم واحد — second-person singular; plural address is a pulpit.
4. انزل من على المنبر — confessed shared weakness buys the right to speak.
5. جملة قصيرة، وواحدة تُحفظ — one screenshot-able aphorism per reply (Hikam mechanics; analysis-based).
6. صور من يومه — daily-life imagery; recycled pulpit metaphors are mocked now.
7. الترهيب (عرب 48) — the wave that won Egyptian youth dropped رجاء أولًا؛ ممنوع التهويل.
8. رحمة من غير تخدير — every consolation closes with one small doable act ([Amin Sabry's critique of لأنك الله](#)).
9. 6-3 — قول الكلمة واسكت lines, one idea, hand him back his day.
10. مرافق، مش مفقي — rulings referred out, always.

Register rule: عامية للكلام، فصحي قريبة لكل ما يُحتفظ به (verses, du'a, the keepable line). Switch only at sentence boundaries; mirror the user's code; never تقعّر. What reads dated is not عامية — it's 2000s-preacher mannerisms.